

Jingzhi Zhang (Jiachi ZHANG)

Tel: +86 15816658985 | Email: jingzhizhang569@gmail.com | GitHub: github.com/tyui99 | tyui99.github.io

RESEARCH INTERESTS

Efficient AI, Model Compression & Acceleration, Medical Image Analysis, Prediction-Space Learning, Brain-Computer Interface

EDUCATION

South China University of Technology | 09/2023–06/2027

B.Eng. in Information Engineering | Average Score: 84.52/100

Core Courses: Signals & Systems, Analog Electronics, Digital Electronics, Digital Signal Processing, Principle of Communications, Microcomputer System and Interface Technology, Programming in C++

École Polytechnique de l'Université de Nantes, France | 07/2025

Summer Study Program | Français, Théorie de l'Information

INTERNSHIP EXPERIENCE

Shenzhen Institutes of Advanced Technology, Chinese Academy of Sciences | 10/2025–Present

Visiting Research Intern | Advisors: Dr. Wenqi Fang & PhD candidate Zhuoyu Wu

- Conducted independent research by identifying research gaps, developing hypotheses, designing experimental protocols, and performing hands-on experiments and data analysis.
- Authored and revised two first-author manuscripts submitted to ICONIP 2026 and APSIPA ASC 2026.
- Compiled academic materials by reviewing relevant literature and drafting concise, structured experimental summaries.

PUBLICATIONS & RESEARCH

Gain-Aware Prediction-Space Recursive Controller for Lightweight Polyp Segmentation

Submitted to ICONIP 2026 | Jingzhi Zhang (Jiachi ZHANG) et al.

- Formulated segmentation refinement as prediction-space recursive correction with only **189 parameters** and negligible GMACs overhead.
- Designed a three-stage recursive controller: Evidence Encoding, Gain-Aware State Update, State-Guided Residual Correction.
- Evaluated across **7 lightweight backbones** on 4 datasets, consistently outperforming training-side baselines (+LL/+DS/+Mutation/+LoMix) and +Harmonizing.

RIGS-Refiner: Risk-Guided Recursive Refinement in Prediction Space for Polyp Segmentation

Submitted to APSIPA ASC 2026 | Jingzhi Zhang (Jiachi ZHANG) et al.

- Proposed a lightweight post-refinement plugin with **519 parameters** combining image priors, prediction cues, risk-guided update, and residual correction.
- Tested on PraNet and SegFormer-B0 across 4 datasets, achieving favorable efficiency–accuracy trade-offs against SegFix, rNCA, and CascadePSP (×2600 parameter reduction).

Glomerular Pathological Image Detection via YOLO11 with Stain Normalization

04/2024–03/2025

- Designed color clustering and transfer modules to mitigate staining variance; built YOLO11-based detection framework with EMA attention and Focal Loss, achieving 94% mAP.

PROJECT EXPERIENCE

Macroeconomic Indicator Forecasting via SAITS and Time-Series LLMs | 07/2025–08/2025

- Built multivariate time-series imputation and forecasting system; integrated TimeLLM and ChatTS with cross-modal alignment. **Top 10** in E Fund FinTech Challenge.

FPGA-Based Shooting Confrontation Game | 03/2026

- Developed dual-mode FPGA game with 1280×800@60fps HDMI output, infrared control, AABB collision detection, and 7-state FSM. **Excellent Course Design Award**.

HONORS & AWARDS

- 3rd Prize Scholarship, South China University of Technology (2025)
- Merit Student, South China University of Technology (2025)
- Advanced Individual in Student Work, SCUT (2025)
- 10th Place, Pazhou Algorithm Competition — E Fund Cup FinTech Challenge (2025)
- Outstanding League Cadre, SCUT (2025)
- Excellent Course Design Award, SCUT (2026)

ADDITIONAL SKILLS

Languages: French (basic), English (fluent, can be used as a studying language)

Programming: Python, PyTorch, Keil, MATLAB, Vivado